

Overcoming the Ambient-Temperature Operation Limitation in Lithium-Ion Batteries by using a Single-Ion Polymer Electrolyte Fabricated by Controllable Molecular Design

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A polymer electrolyte with a particular molecular architecture plays a critical role in the improvement of the performances of both a polymer electrolyte and a battery device. Here, we present the design and synthesis of a sp^3 -boron-based single-ion polymer electrolyte (SIPE), lithium poly(bisphenol AF borate) (LiPFB), by using partially aromatic bisphenol AF as a linker. As a result of the torsional, flexible, and strong electron-withdrawing properties of the bistrifluoromethyl methylene group ($-\text{C}(\text{CF}_3)_2-$), the prepared poly(vinylidene fluoride-co-hexafluoropropene (PVDF-HFP)/LiPFB membrane displays an enhanced ionic conductivity

and a remarkable compatibility with electrodes compared to that of the PVDF-HFP/lithium poly(4,4'-biphenol borate) membrane, in which a linear and fully aromatic 4,4'-biphenol is used as a linker. Consequently, an excellent battery performance both at room temperature and high charge/discharge rates was demonstrated. This study underscores the fundamental importance of the molecular structure of SIPE materials in battery performance enhancement and suggests ways to design new SIPE materials to enhance battery devices with improved electrochemical performance.

Introduction

It has become increasingly evident that current electrolyte technologies cannot be used to develop safe, low-cost, and efficient energy storage technologies with Li-ion batteries because of the drawbacks of the conventional liquid dual-ion electrolytes, LiBF_4 and LiPF_6 , which have a low lithium ion transfer number, poor thermal stability, and sensitivity to moisture.^[1] As a result of the unique advantages of single-ion polymer electrolytes (SIPEs), such as high Li ion transfer number, good thermal and electrochemical stabilities, and excellent mechanical strength, they have been developed widely as a new electrolyte technology in the recent years.^[2–4] The SIPE based on sp^3 boron is one of the most widely investigated SIPEs and has attracted increasing attention because of its excellent ionic conductivity and high thermal and electrochemical stability.^[5,6] Furthermore, sp^3 boron atoms can be bonded covalently to different kinds of linkers to construct sp^3 -boron-based SIPEs with different and desirable performances. For example, linkers with an aromatic benzene ring can facilitate the negative charge of the polymer anion to be redistributed in the extra-framework by aromatic π -conjugation to achieve a reduced dissociation energy of the lithium ion from the polymer anion and an enhanced mobility of Li ions.^[7] Unfortunately, a fully aromatic linker always leads to a highly polar material, which gives rise to the poor room-temperature performance of the cells because of the relatively high interfacial resistance between the SIPE membrane and the electrodes.

Many studies have indicated that the introduction of flexible aliphatic groups into the material can reduce the polarity of the materials significantly.^[8] In the present work, the partially aromatic bisphenol AF linker with a torsional bistrifluoromethyl methylene group ($-\text{C}(\text{CF}_3)_2-$) was used to synthesize the lithium poly(bisphenol AF borate) (LiPFB) SIPE to overcome the room-temperature operation limitation of aromatic-SIPE-based Li-ion batteries. Meanwhile, fully aromatic lithium poly(bisphenol borate) (LiPBB) was synthesized as a reference to investigate the effect of the torsional $-\text{C}(\text{CF}_3)_2-$ group on the physicochemical properties of the LiPFB and the performance of the fabricated Li-ion batteries. We used SEM and XRD to confirm that the torsional $-\text{C}(\text{CF}_3)_2-$ group plays a vital role to reduce the polarity of the LiPFB by decreasing its ordered structure. A poly(vinylidene fluoride-co-hexafluoropropene (PVDF-HFP)/LiPFB membrane was then prepared by the solution casting method. As a result of the strong electron-withdrawing effect

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of the $-\text{C}(\text{CF}_3)_2-$ group, the PVDF-HFP/LiPFB membrane displays a two-times-higher ionic conductivity than the PVDF-HFP/LiPBB membrane. In particular, a low interfacial resistance of $72\ \Omega$ was obtained for the PVDF-HFP/LiPFB membrane Li-ion battery, which is much lower than the value of $407\ \Omega$ for the PVDF-HFP/LiPBB membrane Li-ion battery. As a result of the combination of the high ionic conductivity and low interfacial resistance, good room-temperature performance of the PVDF-HFP/LiPFB membrane Li-ion battery was achieved. We believe this paves the way for the design and elaboration of a new class of SIPEs toward the practical application of the SIPE-based Li-ion batteries.

Results and Discussion

Chemical structure

It is apparent from the molecular structures of the two SIPE materials that LiPFB should display a random disordered structure because of the torsional and flexible $-\text{C}(\text{CF}_3)_2-$ group, as compared to the regular and rigid structure of LiPBB with the linear 4,4'-biphenol group (Scheme 1). To confirm this, we performed field-emission scanning electron microscopy (FE-SEM) and XRD to investigate their microstructures. LiPFB shows a disordered structure in the FE-SEM image (Figure 1 a) and in the broad peaks of the XRD pattern (Figure 1 a') with an amorphous morphology and microstructure. In contrast, LiPBB shows a cube-like shape with good crystallinity (Figure 1 b, b').^[4,9,10] The disorder and random microstructure demonstrates that the LiPFB displays a low polarity, which is important to increase the flexibility of the PVDF-HFP/LiPFB membrane and enhance the compatibility of the polymer electrolyte with the electrode. In contrast, the high polarity of LiPBB that arises from the linear aromatic diphenol linkers will lead to the high rigidity of the PVDF-HFP/LiPBB membrane, which gives rise to

poor compatibility with the electrode and high interfacial resistance. Apparently, the low interfacial resistance favors an enhancement in the battery performance.^[11]

Morphology

The porosity of polymer electrolytes plays a vital role in improving the ionic conductivity of the polymer electrolyte and the performance of Li-ion batteries.^[12] The morphology of the SIPE membranes was investigated by using FE-SEM (Figure 2). Both membranes display smooth morphologies with very small blind pores both on the membrane surfaces and cross-sectional images. However, the PVDF-HFP/LiPFB membrane demonstrates a more porous structure than the PVDF-HFP/LiPBB membrane mainly because of the poor compatibility between the flexible aliphatic PVDF-HFP and the rigid fully aromatic LiPBB.^[10]

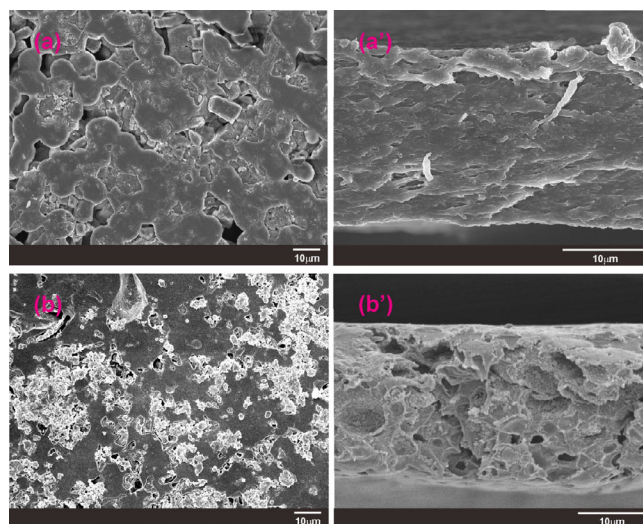


Figure 2. SEM images of a) LiPFB and b) LiPBB surfaces and a') LiPFB and b') LiPBB cross-sections.

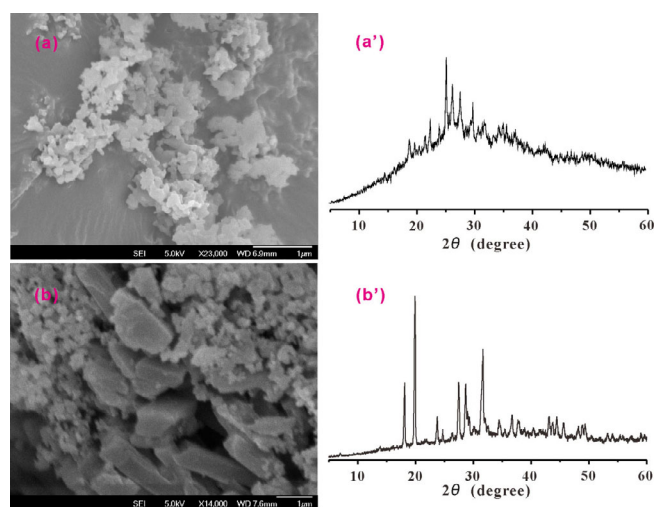


Figure 1. SEM images of a) LiPFB and b) LiPBB and XRD patterns of a') LiPFB and b') LiPBB.

Ionic conductivity

Indeed, in addition to the flexibility of the $-\text{C}(\text{CF}_3)_2-$ group, the influence of the strong electron-withdrawing effect of the $-\text{C}(\text{CF}_3)_2-$ group on the ionic conductivity should be taken into account. The calculated dissociation energies and the calculated electrostatic potentials of LiPFB, LiPBB, and lithium poly(bisphenol A borate) (LiBPAB) are compared in Figure 3.^[13] Clearly, the value of the dissociation energy directly reflects the difficulty of the dissociation of the Li ion from the polymer anion. The dissociation of the Li cation from LiPFB requires $100.9\ \text{kcal mol}^{-1}$, which is much lower than the values of $112.8\ \text{kcal mol}^{-1}$ for LiPBB and $115.3\ \text{kcal mol}^{-1}$ for LiBPAB (Figure 3a). The results indicate that LiPFB exhibits the most efficient Li ion transport because it has the weakest interaction between the Li cation and poly-(bisphenol AF borate) anion, which gives rise to a high ion conductivity. To the best of our knowledge, the difference of

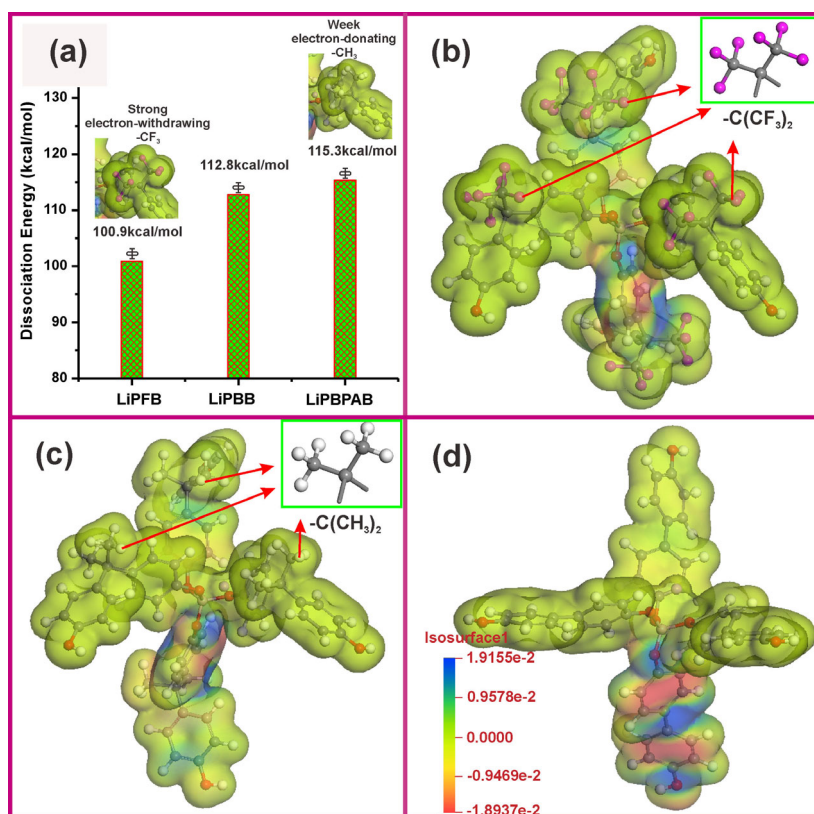


Figure 3. a) Calculated dissociation energies of LiPFB, LiPBB, and LiBPAB and the calculated electrostatic potentials of b) LiPFB, c) LiBPAB, and d) LiPBB.

the dissociation energies among the three materials is attributed to the electron-withdrawing effect of the $-\text{C}(\text{CF}_3)_3$ -group of LiPFB (Figure 3b) and the weak electron-donating effect of the $-\text{C}(\text{CF}_3)_3$ -group of LiBPAB (Figure 3c) compared to LiPBB, without any electron-withdrawing or electron-donating groups (Figure 3d).^[14,15] Moreover, the electrostatic potentials indicate that the anionic charges that reside on the B atoms of all three materials are indeed delocalized in its vicinity along the aromatic benzene rings in the polymers, which enhances the ionic conductivity of the SIPE membrane significantly. The PVDF-HFP/LiPFB membrane displays a high ionic conductivity of 0.39 mS cm^{-1} at 25°C and 1.88 mS cm^{-1} at 80°C , respectively (Figure 4), which are almost two times higher than that of the PVDF-HFP/LiPBB membrane.^[10] Furthermore, the activation energies were calculated to be 25.27 and $19.86 \text{ kJ mol}^{-1}$ for the PVDF-HFP/LiPFB and PVDF-HFP/LiPBB membranes, respectively, which is modest among the reported polymer electrolytes.^[16,17] Notably, the PVDF-HFP/LiPFB membrane demonstrates a slightly higher activation energy than the PVDF-HFP/LiPBB membrane, which might be derived from the low Li ion condensation and the long bisphenol AF linker of LiPFB.

Interfacial resistance

The electrochemical impedance spectroscopy (EIS) responses of the Li-ion batteries assembled with the SIPE mem-

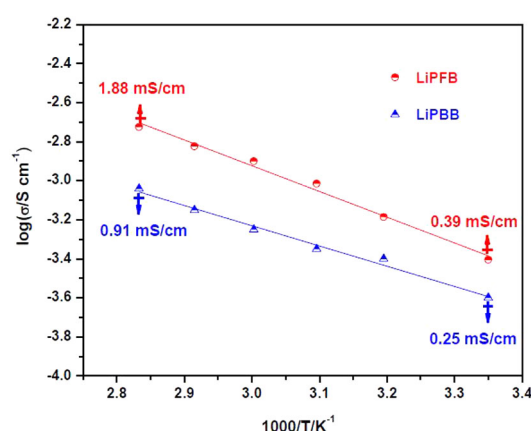


Figure 4. Temperature dependence of the ionic conductivities of the PVDF-HFP/LiPFB and the PVDF-HFP/LiPBB membranes.

branes at room temperature in the Nyquist coordinates are depicted in Figure 5a,b. The equivalent circuit performed to fit the EIS results is also provided, in which R_e , R_i , and R_{ct} are the bulk resistance, interfacial resistance, and charge transfer resistance, respectively. CPE1, CPE2, and CPE3 are the constant phase elements of the membrane and the electrical double layer (EDL) at the both sides of the membrane, respectively. The Li|PVDF-HFP/LiPFB membrane|LiFePO₄ cell displays a remarkably low interfacial resistance of 72Ω , which is comparable to that of a commercial liquid electrolyte Li-ion battery.^[18] In contrast, a high interfacial resistance

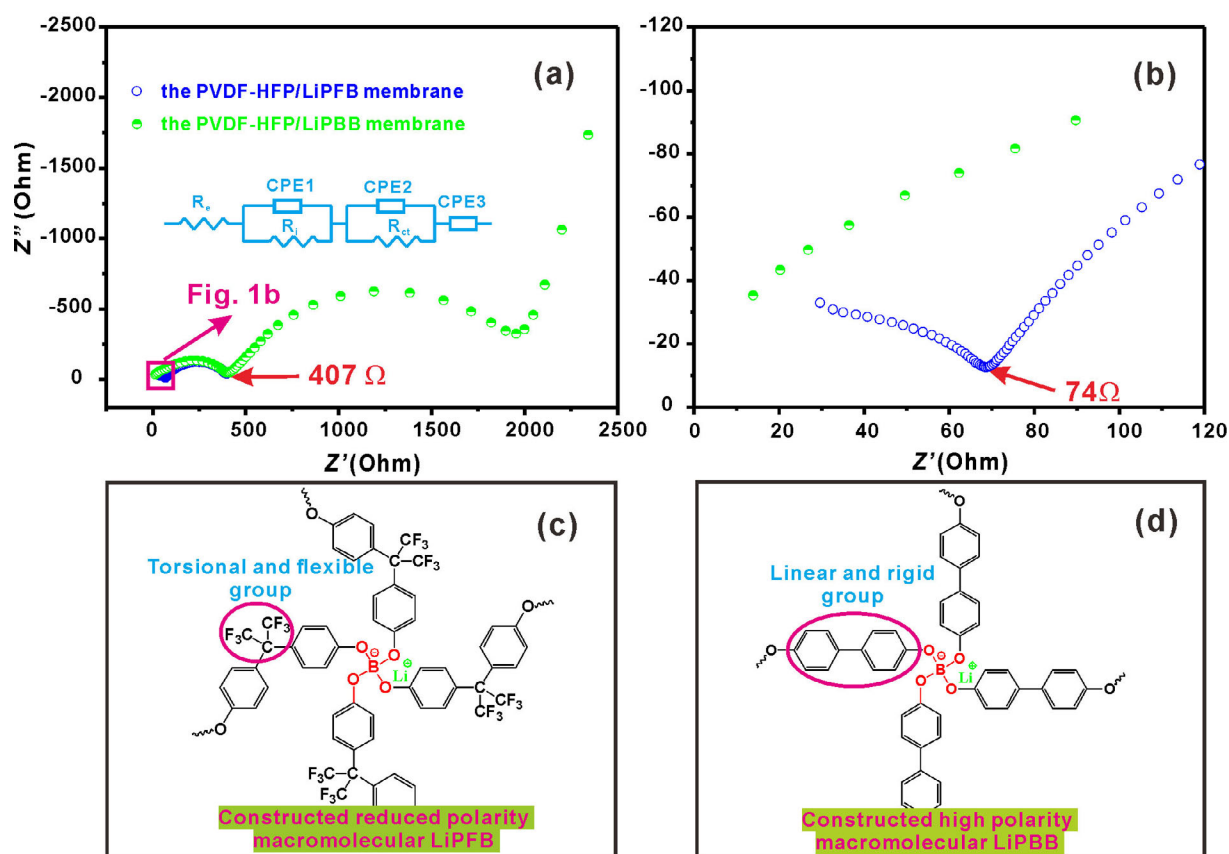


Figure 5. a) Nyquist plots of typical Li|SIPE membrane|LiFePO₄ cells in the full frequency range and b) Nyquist plots of typical Li|SIPE membrane|LiFePO₄ cells in the selected frequency range. Chemical structures of c) LiPF₆ and d) LiPBB.

of 407 Ω was obtained for the Li|PVDF-HFP/LiPBB membrane|LiFePO₄ cell.^[10] A comparison of the chemical structures of LiPF₆ and LiPBB indicates clearly that the $-\text{C}(\text{CF}_3)_2-$ group plays a vital role to reduce interfacial resistance by enhancing the flexibility of the PVDF-HFP/LiPF₆ membrane caused by low polarity, which increases the compatibility with the electrodes.

Battery performance

The room-temperature performance shows high potential for practical application in Li-ion batteries. Great efforts have been made by using different approaches such as polymer composites and blends,^[19] porous polymer electrolyte membranes,^[8] and chemical structure design^[3,6,15] for the room-temperature operation of SIPE-based Li-ion batteries with high performance. Although the room-temperature operation of SIPE-based Li-ion batteries has been achieved previously, a low discharge capacity at a high C-rate still limits its practical application.^[20,21] In the present work, the performance of the Li|PVDF-HFP/LiPF₆ membrane|LiFePO₄ device at room temperature was achieved because of the improved ionic conductivity and interfacial resistance. The rate capabilities of the Li|PVDF-HFP/LiPF₆ membrane|LiFePO₄ device are demonstrated in Figure 6a,b. The Li|PVDF-HFP/LiPF₆ membrane|LiFePO₄ cell exhibits ex-

cellent room-temperature performance with average discharge capacities of 133 and 97 mAh g⁻¹ at high discharge rates of 0.1 and 2 C, respectively. The battery performance of the PVDF-HFP/LiPF₆ membrane is particularly remarkable because the SIPE membranes can be operated at room temperature. In addition, the PVDF-HFP/LiPF₆ membrane exhibits an outstanding electrochemical stability, and the discharge capacity of the cell declines only marginally. Moreover, the Coulombic efficiency still remains above 99% after 150 cycles at 2 C (Figure 6c). Unfortunately, the battery assembled with the PVDF-HFP/LiPBB membrane was not operational at room temperature because of its low ionic conductivity and high interfacial resistance. Furthermore, the excellent device performance achieved with the PVDF-HFP/LiPF₆ membrane is also attributed to its high thermal stability at 150 °C (Figure S2, Supporting Information), good electrochemical stability at 5.0 V (Figure S3, Supporting Information), and excellent mechanical strength at 7.2 MPa (Figure S4, Supporting Information).^[17] In particular, the Li ion transfer number (t_{Li}^+) of LiPF₆ was close to unity (Figure S5 and Table S1, Supporting Information), which is substantially higher than that of conventional liquid electrolytes (<0.3).^[21,22] The significantly higher value of t_{Li}^+ confirms that LiPF₆ indeed exhibits single-ion behavior because of the restricted movement of the anions attached to the polymer backbone.^[23]

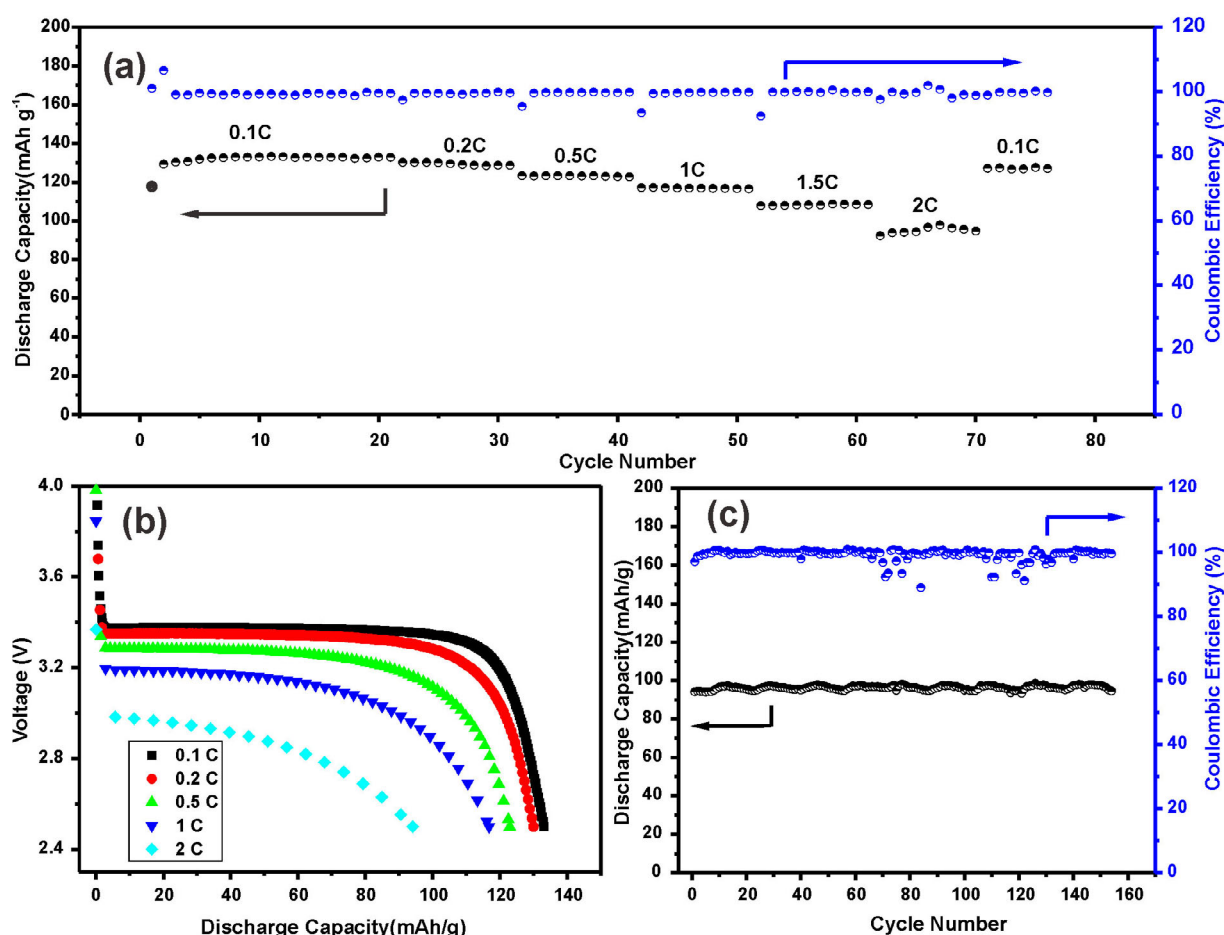


Figure 6. a) Rate capability of the Li|PVDF-HFP/LiPFB membrane|LiFePO₄ cells at 25 °C; b) discharge voltage profiles of the Li|PVDF-HFP/LiPFB membrane|LiFePO₄ cells at 25 °C; c) cycling performance of the Li|PVDF-HFP/LiPFB membrane|LiFePO₄ cells at 2 C and 25 °C.

Conclusions

We report a lithium poly(bisphenol AF borate) material as a single-ion polymer electrolyte (SIPE) with unprecedented room-temperature performance for SIPE-based lithium-ion batteries in terms of its high ionic conductivity (above 10^{-4} S cm⁻¹), remarkably improved interfacial resistance 72 Ω compared with the value of 407 Ω for a lithium-ion battery based on lithium poly(bisphenol borate) material, and the excellent electrochemical stability window of 4.7 V versus Li⁺/Li. We believe that the promising properties of lithium poly(bisphenol AF borate) are caused by the introduction of the flexible and strong electron-withdrawing $-C(CF_3)_2-$ group. This implies that the design and optimization of a single-ion polymer electrolyte can enable SIPEs to meet the requirements for future lithium-ion battery technology in electric transport applications.

Experimental Section

Materials

Bisphenol, bisphenol AF, trimethylborate, PVDF-HFP, LiFePO₄, and acetylene black were purchased from Sigma-Aldrich. Aceto-

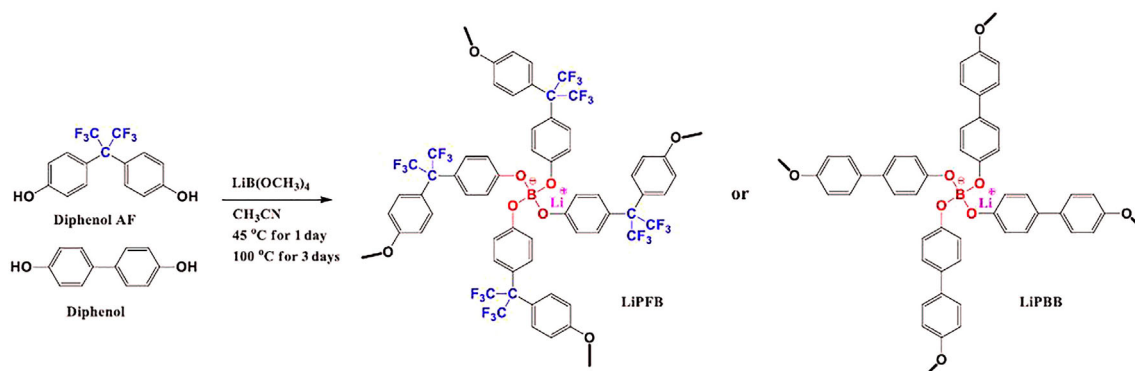
nitrile was dried over phosphorus(V) oxide and distilled. Methanol was dried with magnesium and distilled. Bisphenol AF was dried under vacuum at 80 °C overnight before use.

Synthesis of lithium tetramethanolatoborate

Lithium tetramethanolatoborate [LiB(OCH₃)₄] was synthesized and purified by a method described by Barthel et al.^[4] Li (1.27 g) was added into a flask under Ar. Methanol (30 mL) was then added dropwise into the flask over 30 min. After that, another portion of methanol (40 mL) was added. The resultant mixture was heated to 60 °C to form a homogeneous solution. Boric acid trimethyl ester (20 mL) was added slowly to the solution, and the reaction was heated for 2 h. Finally, the solution was left at RT for another 24 h. The product was collected by filtration, washed with methanol, and dried.

Synthesis of LiPFB and LiPBB

The syntheses of LiPFB and LiPBB are shown in Scheme 1. The calculated amounts of the linkers, diphenol or diphenol AF, and LiB(OCH₃)₄ were dispersed in acetonitrile by stirring under a stable Ar flow. The mixtures were heated to reflux at 45 °C for 24 h and then at 100 °C for another 72 h. After the mixtures were cooled, the raw products were collected by filtration and further purified by acetonitrile at 90 °C for 6 h. The final products were



Scheme 1. Synthesis of LiPFB and LiPBB.

dried under vacuum at 100 °C for 24 h. The obtained materials were stored in an Ar-filled glovebox. ^{11}B NMR and ^1H NMR spectra confirmed that the SIPE was synthesized successfully (Figure S1, Supporting Information).

Preparation of SIPE membranes

To fabricate PVDF-HFP/SIPE membranes, PVDF-HFP and LiPFB or LiPBB at a weight ratio of 3:1 were dissolved in DMF at 80 °C to form homogeneous solutions. The resultant solutions were subsequently cast onto glass Petri dishes and kept in an oven at 80 °C to evaporate the DMF. The SIPE membranes were peeled off and dried at 80 °C under vacuum for 48 h. Finally, the fabricated SIPE membranes were kept in an Ar-filled glovebox and soaked in ethylene carbonate/propylene carbonate (EC/PC; 1:1 v/v) before they were characterized. The membrane thickness was approximately 25 μm .

Preparation of LiFePO_4 cathode

The cathode film was prepared by casting a well-stirred *N*-methyl-2-pyrrolidone (NMP) solution composed of LiFePO_4 (70 wt %), acetylene black (20 wt %), and PVDF (10 wt %) on an Al foil. The resulting electrode was dried at 60 °C for 6 h and at 60 °C under vacuum for 12 h. The well-dried cathode was cut into circular pieces for use in coin cells. Standard coin cells (C1625) were assembled in a glovebox. The loading density of LiFePO_4 was approximately 4 mg cm^{-2} . The C rate is often used to describe battery charging, and 1 C was the capacity rating [Ah] of the battery (1 C = 120 mAh g^{-1}).

Characterization

^1H NMR spectra were recorded by using a Bruker Ascend TM at 400 MHz. $[\text{D}_6]\text{DMSO}$ was used as the solvent for the ^1H NMR measurements. ^{11}B NMR was performed by using a Bruker Ascend TM at 400 MHz using boron (tri)fluoride etherate as a solvent. The morphologies of the polymer electrolytes both on the surface and in cross-section were probed by using SEM (SU8010). Samples were sputtered with Pt before measurement under 5×10^{-2} mbar at 25 °C by using a BaltecSCD050 apparatus (120 s, 30 mA). The thermal stability was measured under a stable N_2 flow (60 $\text{cm}^3 \text{min}^{-1}$) with a 10 °C min^{-1} heating rate by using a thermogravimetric analyzer (STA 409 PC). The thermal stability test was conducted over the range of 30–600 °C. XRD

patterns were recorded from $2\theta = 5$ to 60° by using a Bruker D8-Focus with scan speed of 0.02° s^{-1} . The ion conductivity of the polymer electrolytes was measured by using EIS by using a Zahner potentiostat–galvanostat electrochemical workstation model with a frequency range from 1×10^6 to 1 Hz and an oscillating voltage of 5 mV. A cylindrical stainless-steel device with a diameter of 1.5 cm was used for conductivity testing. The raw data was fitted using the simulated impedance measurement (SIM) software. The electrochemical stability test (cyclic voltammetry) was performed by using the same stainless-steel device from –0.4 to 7 V by using a CHI instrument at a scan rate of 1 mV s^{-1} by using the Li metal as a reference. The Li ion transfer number (t^+) was calculated by the equation: $t^+ = I_s(\Delta V - I_0 R_0) / I_0(\Delta V - I_s R_s)$, in which ΔV is the potential applied to the cell, I_0 and I_s are the initial and steady-state currents, respectively, and R_0 and R_s are the initial and steady-state resistances of the passivation layers on the Li electrode, respectively, measured from the Li|SIPE membrane|Li battery cell. The SIPE membrane was soaked in an EC/PC solution before measurement.

Computational details

Density functional theory (DFT) was employed to calculate the electronic structure under the generalized gradient approximation (GGA).^[24] The electronic structure calculations were performed with a self-consistent-field (SCF) tolerance of 10^{-6} eV using Perdew–Wang’s exchange–correlation functional (PW91)^[25] as implemented in the Vienna ab initio simulation package (VASP) code.^[26] The dissociation energies and the electrostatic potentials were calculated by DFT under GGA, and the values were calculated by the following equation: $E_d = E_{\text{poly}^-} + E_{\text{Li}^+} - E_{\text{Li-poly}}$, in which E_{poly^-} and E_{Li^+} are the energies of the polymeric anion and Li cation, respectively. $E_{\text{Li-poly}}$ represents the energies of polymer electrolytes.^[27]

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Conflict of interest

The authors declare no conflict of interest.

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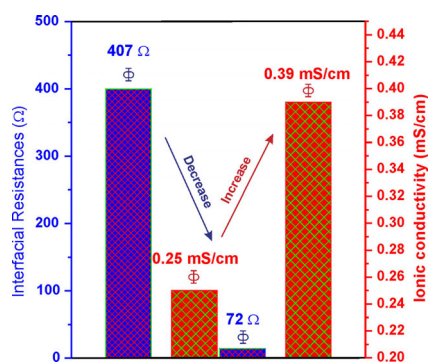
FULL PAPERS

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Overcoming the Ambient-Temperature Operation Limitation in Lithium-Ion Batteries by using a Single-Ion Polymer Electrolyte Fabricated by Controllable Molecular Design



Turn on the electrolyte: The electrolyte interfacial resistance and ionic conductivity are enhanced by using a controllable molecular design fabricate a high-performance single-ion polymer electrolyte to overcome the ambient-temperature operation limitations in lithium-ion batteries.